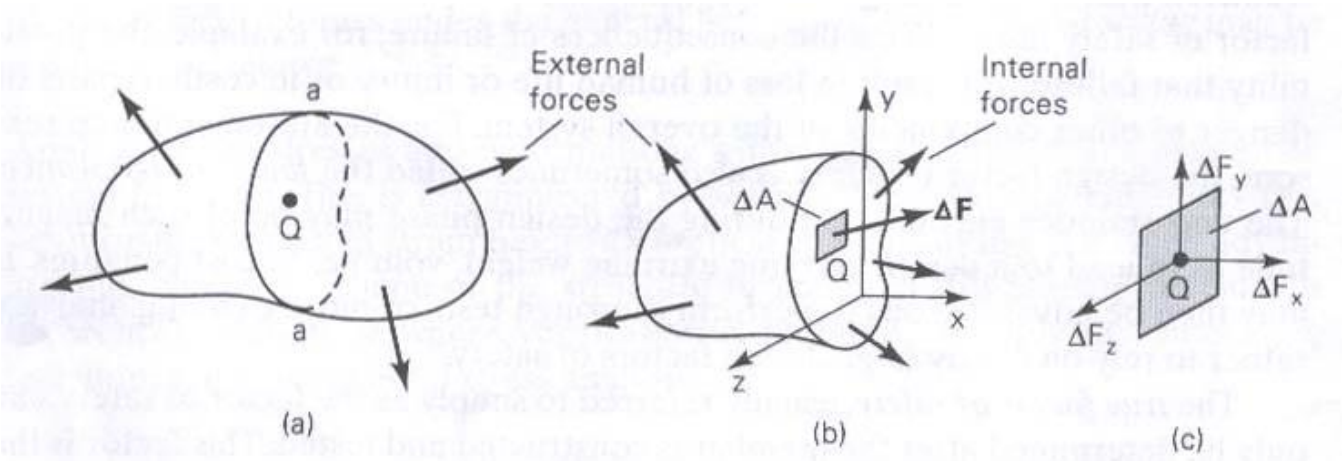
Chap. 2 Stress

2.1 Definition of stress

- Under the action of external forces, internal forces will be developed within the body.



▼ As the forces acting on <u>the entire body are in equilibrium,</u> the forces acting on one part <u>alone must be in equilibrium</u>

=> These internal forces, applied to both parts, are distributed continuously over the cut surface.

• Consider an element of area ΔA, located at point Q on the cut surface is acted on by force ΔF.

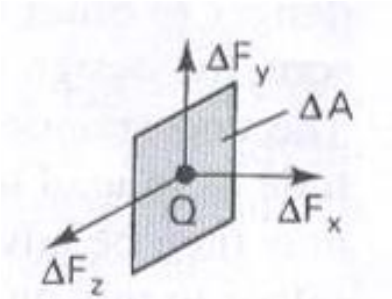
Let the origin of coordinates be placed at point Q.
=> Decomposing ΔF into components parallel to x, y, z.

▼ We define the normal & shearing stresses :

$$\left(\begin{aligned} \sigma_x &= \lim_{\Delta A \to 0} \frac{\Delta F_x}{\Delta A} = \frac{dF_x}{dA} \\ \tau_{xy} &= \lim_{\Delta A \to 0} \frac{\Delta F_y}{\Delta A} = \frac{dF_y}{dA} \\ \tau_{xz} &= \lim_{\Delta A \to 0} \frac{\Delta F_z}{\Delta A} = \frac{dF_z}{dA} \end{aligned}\right)$$

normal stress

shear stress

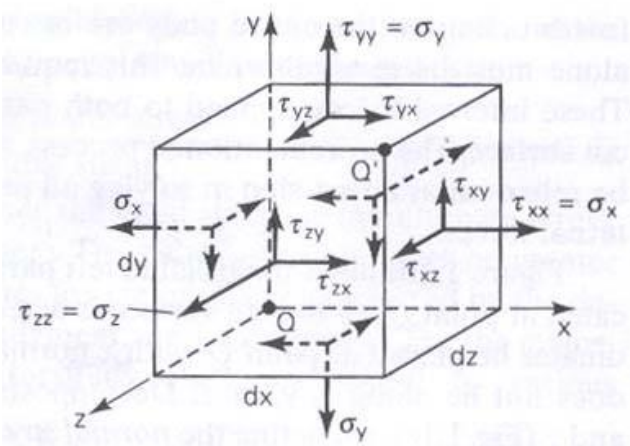


(c)

depends on orientation of ΔA *as well as* $\Delta \underset{\sim}{F}$

▼ Very small area but larger than interatomic distance scale.
=> Stress will be changed depending on the selection of area.

2.2 Components of stress: stress tensor



• Consider the stresses to be identical at point Q and Q' and uniformly distributed on each face, the matrix form of stress is

$$[\tau_{ij}] = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

τ_{ij} → The direction of the force

The direction of a normal to the plane or face on which the stress component acts

Sign convention

- When both the outer normal vector of surface and the stress component show a positive direction, the stress is positive. Also, when both the outer normal vector and the stress component show a negative direction, the stress is positive.

Equality of shearing stresses

for equilibrium of the cubic element.

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum F_z = 0, \quad \& \sum M_x = 0, \ldots$$

$$\sum M_z = 0$$

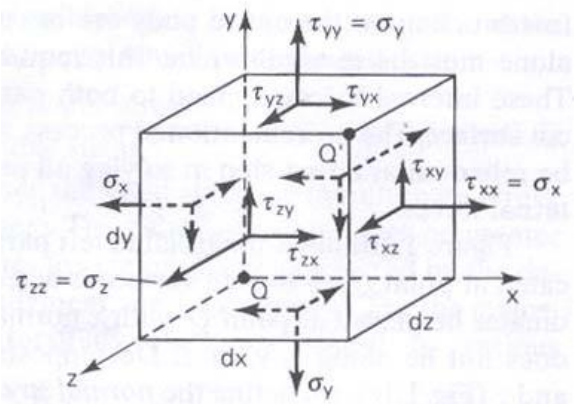
$$\rightarrow \left(-\tau_{xy}\, dydz\right)dx + \left(\tau_{yx}\, dxdz\right)dy = 0$$

$$\tau_{xy} = \tau_{yx}$$

likewise, $\sum M_y = 0 \rightarrow \tau_{xz} = \tau_{zx}$

$$\sum M_x = 0 \rightarrow \tau_{xy} = \tau_{yx}$$

symmetric τ_{ij}



Indicial Notation

For more compact expression for stress, strain, etc., tensor notation is used.

- Vector is first-order tensor, u_i
- Stress, strain are second-order tensor, $\sigma_{ij}, \varepsilon_{ij}$
- Mass, temperature are scalar, m, T

- Two simple conventions

a. Range convention – When a lowercase alphabetic subscript is unrepeated, it takes on all values indicated.

b. Summation convention – ~ is repeated in a term, then summation over the range of that subscript is indicated.

$$\tau_{ij,j} + F_i = 0, \quad i, j = x, y, z \leftrightarrow Eq.(1.11)$$

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$$

⇒ The equations of transformation of the components of a stress tensor.

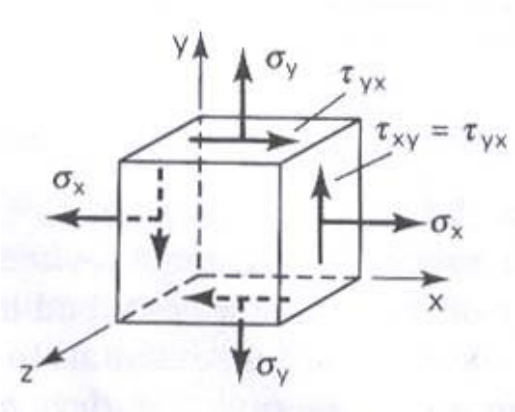
$$\tau'_{rs} = l_{ir} l_{js} \tau_{ij} \qquad (see\ Eq.(1.24))$$

2.3 Some special cases of stress

- Triaxial stress
  Absence of shearing stresses → principal stress.

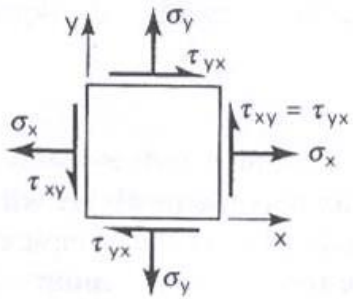
$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

If all principal stresses are equal → Hydrostatic stress.



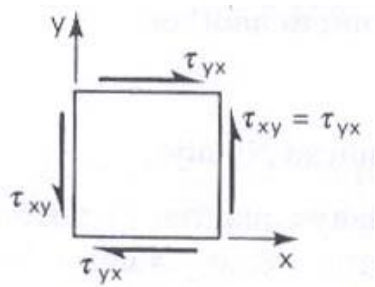
- Two-dimensional or plane stress

$$\begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix}$$



When only two normal stresses are present, the state of stress is called biaxial → occur in thin plates stressed in two mutually perpendicular directions.

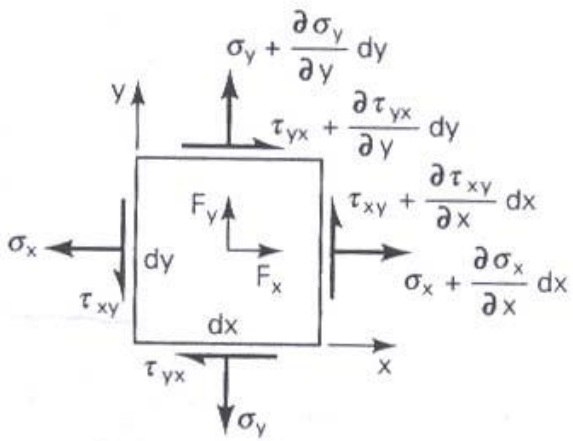
- Pure Shear
  The element is subjected to plane shearing stresses only,
  pure shear occurs over the cross sections and
  on longitudinal planes of a circular shaft subjected to torsion.



- Uniaxial Stress
  When normal stresses act along one direction only
  → the stress of the one-dimensional state.
  → a uniaxial tension or compression.

2.4 Variation of stress within a body

- Consider a thin element of sides dx and dy.



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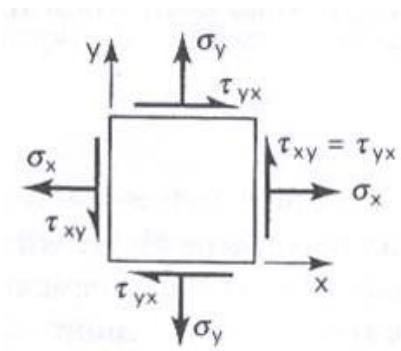
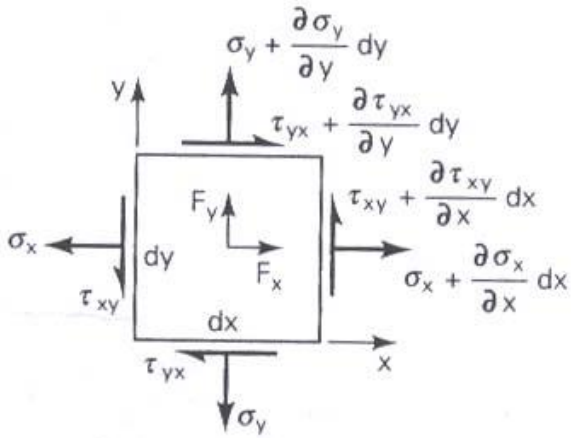


① Assume that σ_x, σ_y, τ_{xy} and τ_{yx} are functions of x, y but do not vary throughout the thickness & $\sigma_z=\tau_{xz}=\tau_{yz}=0$.

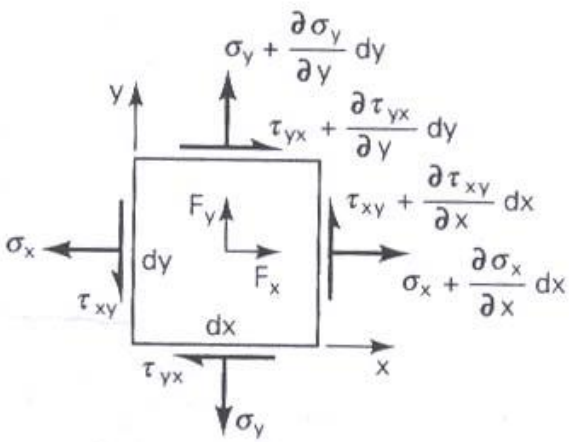
② Assume that the x and y components of the body forces per unit volume, F_x and F_y are independent of z-coord and $F_z=0$.

③ Note that because the element is very small, the stress components may be considered to be distributed uniformly over each face (Should be a single vector representing the mean values applied at the center of each face).



- As we move from one point to another, <u>the variation of stress</u> with position may be expressed by a truncated Taylor's expansion.

$$\sigma_x + \frac{\partial \sigma_x}{\partial x} dx$$

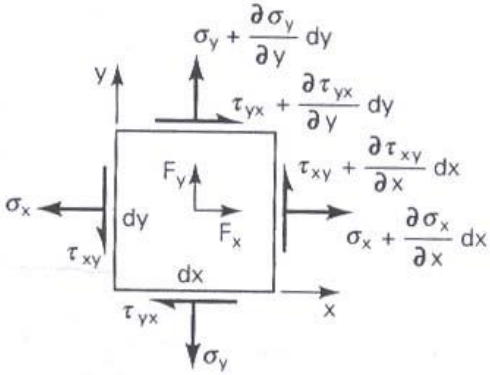


- Consider $\Sigma M_z = 0$ yields.

$$\sum M_z = 0$$

$$\left(2\tau_{xy} + \frac{\partial \tau_{xy}}{\partial x}dx\right)dydz\frac{dx}{2} - \left(2\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y}dy\right)dxdz\frac{dy}{2} = 0$$

$$\rightarrow \tau_{xy} = \tau_{yx}$$



- $\Sigma F = 0$ yields.

$$\sum F_x = 0$$

$$\left(\sigma_x + \frac{\partial \sigma_x}{\partial x}dx\right)dydz - \sigma_x dydz + \left(\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y}dy\right)dxdz$$

$$-\tau_{yx}dxdz + F_x dxdydz = 0$$

$$\rightarrow \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + F_x = 0$$

$$\sum F_y = 0, \quad likewise. \quad \rightarrow \quad \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + F_y = 0$$

for 3 – dimensional stress

$$\left.\begin{aligned} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + F_x = 0 \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + F_y = 0 \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + F_z = 0 \end{aligned}\right] \qquad \underline{\tau_{ij,j} + F_i = 0}$$

3 eqs. with 6 unknown stresses

$\rightarrow$ internally statically indeterminate

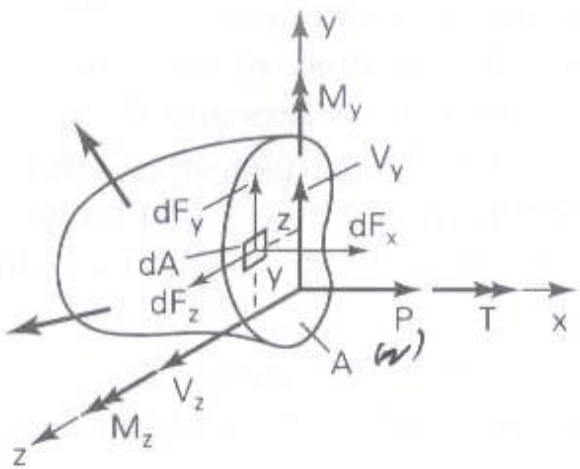
In practical applications,

$$F_x = F_z = 0, \quad F_y = -\rho g << \tau_{ij} \rightarrow \text{ignored}$$

cf dynamic system.

2.5 Internal force-resultant and stress relations

- If there exist external force, distributed forces within a member are presented
  → Stresses can be represented by a statically equivalent system consisting of a force and a moment vector acting at any arbitrary point.



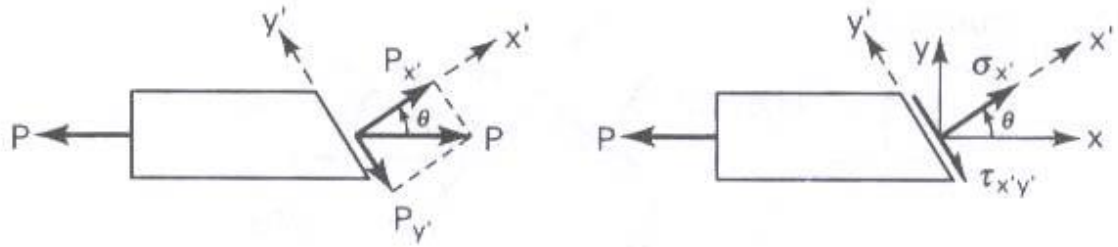
$$\left[\begin{array}{ll} P \ \ or \ \ N & \Rightarrow P = \int \sigma_x dA, \\ V_y, \ \ V_z & \Rightarrow V_y = \int \tau_{xy} dA, \ \ V_z = \int \tau_{xz} dA \\ T & \Rightarrow \ \ T = \int \left(\tau_{xz} y - \tau_{xy} z\right) dA \\ M_y, \ M_z & \Rightarrow M_y = \int \sigma_x z dA, \ \ M_z = -\int \sigma_x y dA \ \ (1.6b) \end{array}\right] \qquad (1.6a)$$

- Other elementary formulas for stress and deformations
  → the basic formulas of mechanics of materials.
  See Table.1.1

  ✓ Derive σ_x=(−) My/I, τ=Tρ/J using a part of beam and bar, respectively

▼ When a member is acted on simultaneously by two or more load types, various internal force-resultants are caused on a section.
  → The final or combined stress is then determined by superposition of the several states of stress.

2.6 Stress on inclined planes under one-dimensional load



- The stresses on an inclined plane

$$P \begin{cases} P_{x'} = P\cos\theta \\ P_{y'} = -P\sin\theta \end{cases} \quad \& \quad A_{x'} = \frac{A}{\cos\theta}$$

$$\therefore \sigma_{x'} = \frac{P\cos\theta}{A_{x'}} = \sigma_x \cos^2\theta$$

$$\tau_{x'y'} = -\frac{P\sin\theta}{A_{x'}} = -\sigma_x \sin\theta\cos\theta$$

- Previous Eqns. indicate how the stresses vary as the inclined plane is cut at various angles.

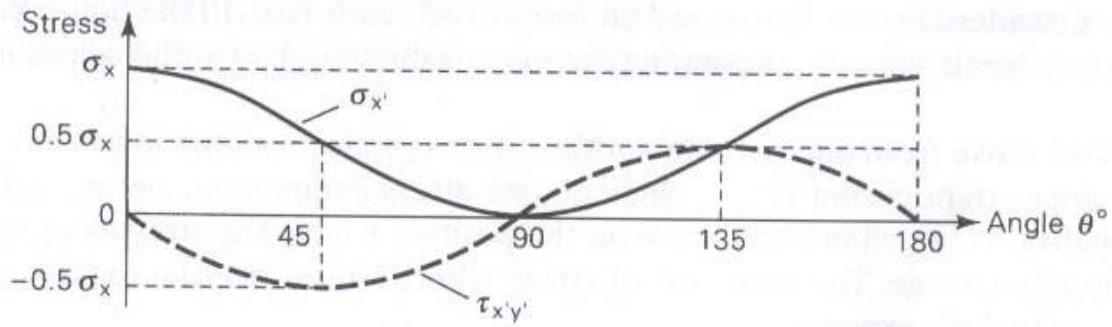


FIGURE 1.7. *Variation of stress at a point with the inclined section in the bar shown in Fig. 1.5.*

$$\therefore \sigma_{x'} = \frac{P\cos\theta}{A_{x'}} = \sigma_x \cos^2\theta$$

$$\tau_{x'y'} = -\frac{P\sin\theta}{A_{x'}} = -\sigma_x \sin\theta\cos\theta$$

▼ The shear stress is zero when the normal stress is either maximum or minimum (See Fig.1.7)

$$\therefore \sigma_{x'} = \frac{P\cos\theta}{A_{x'}} = \sigma_x \cos^2\theta$$

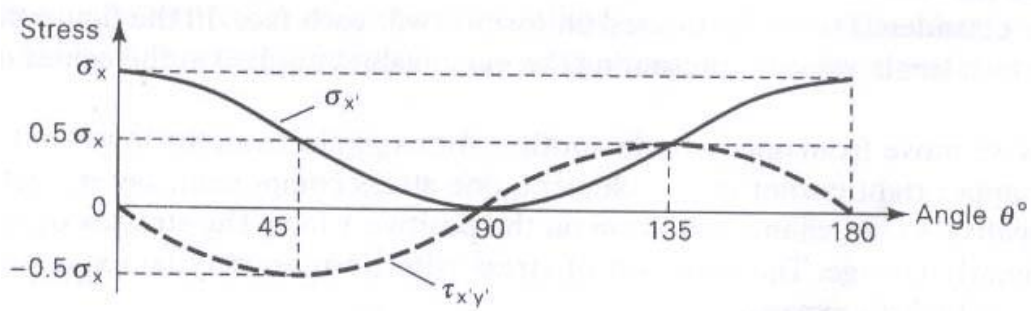
$$\tau_{x'y'} = -\frac{P\sin\theta}{A_{x'}} = -\sigma_x \sin\theta\cos\theta$$

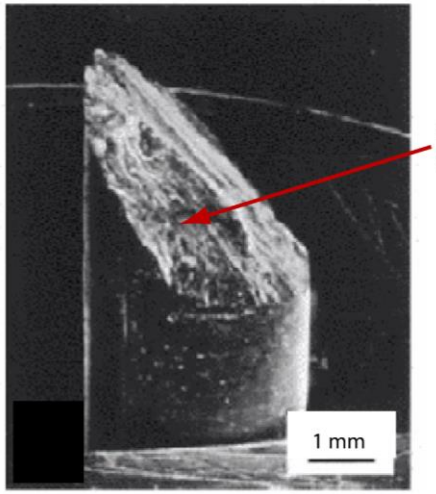
when θ is 0° or 180°, $\sigma_{x'} = \sigma_{max} = \sigma_x$

θ is 45° or 135°, $\tau_{x'y'} = \tau_{max} = \pm\frac{1}{2}\sigma_x$

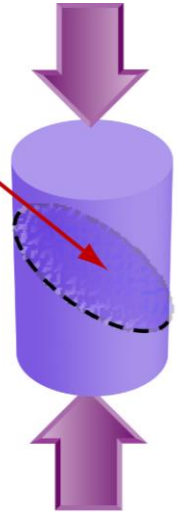
① θ > 90° the sign of $\tau_{x'y'}$ changes.

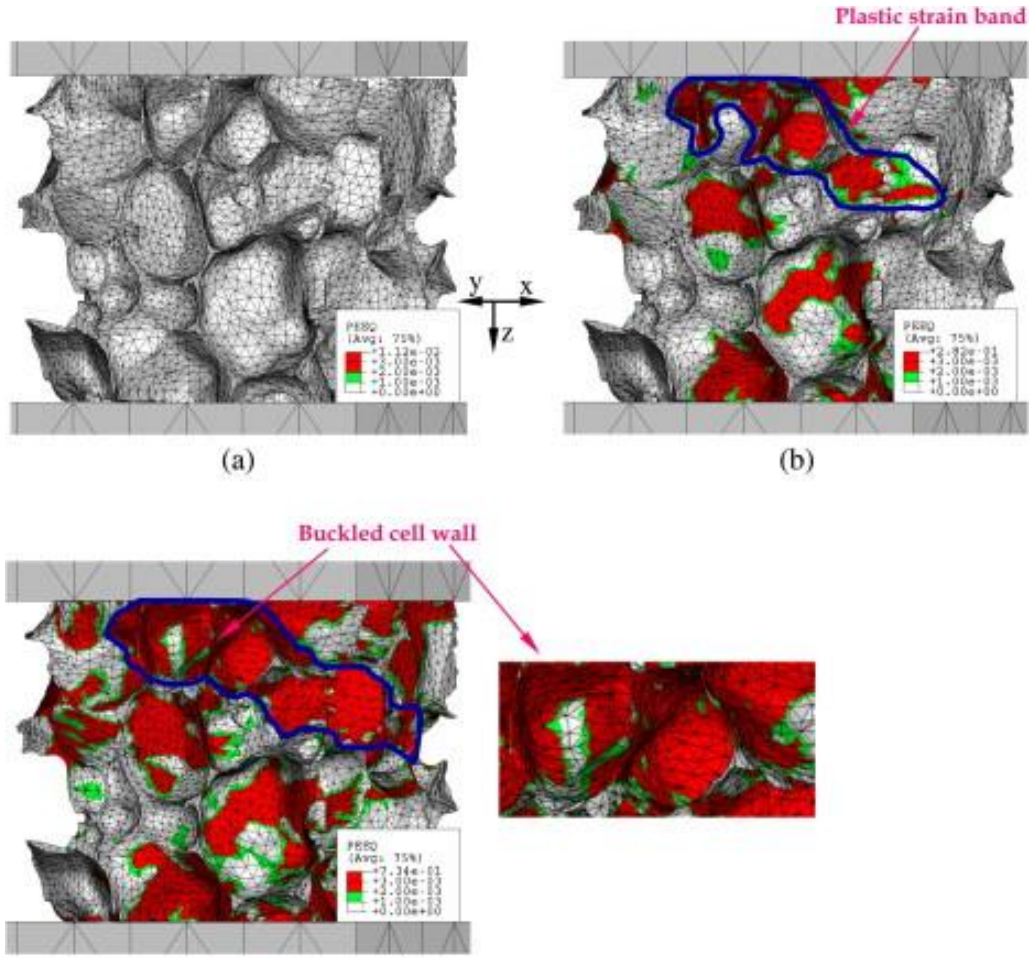
② The magnitude of the shearing stress for any angle θ is equal to that for θ + 90° (opposite sign)





failure plane
or shear band





2.7 Two-dimensional stress at a point

- Two-dimensional state of stress

i) Plane stress

$$\sigma_z = \tau_{xz} = \tau_{yz} = 0, \qquad F_z = 0$$

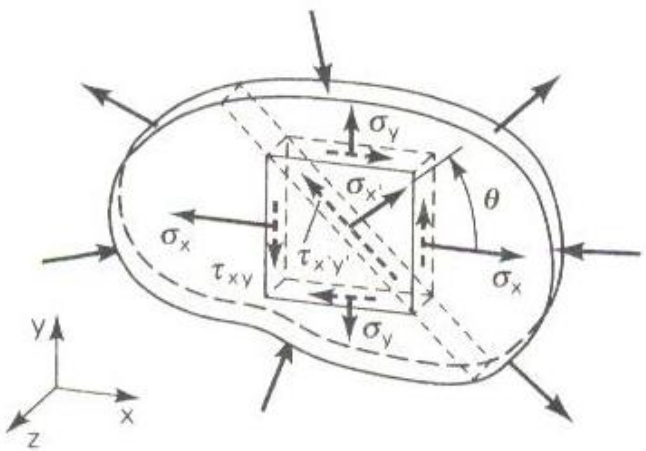
ii) Plane strain

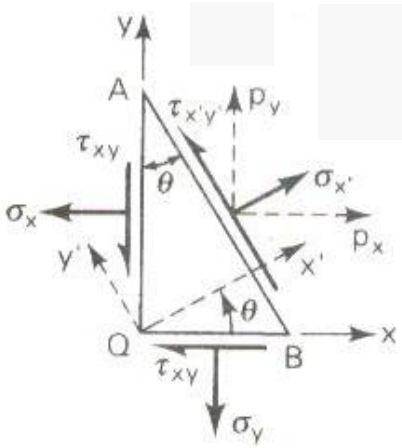
$$\tau_{xz} = \tau_{yz} = 0, \qquad F_z = 0$$

$$\sigma_z = f(\sigma_x, \sigma_y), \qquad \varepsilon_z = 0$$

- The equations for transformation of the stress components, σ_x, σ_y, τ_{xy}

  → consider an infinitesimal wedge cut from the loaded body.



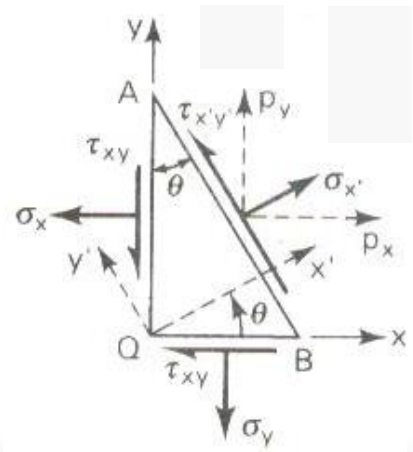


From equilibrium of forces in the x & y directions (AB is unit length).

$$p_x = \sigma_x \cos\theta + \tau_{xy} \sin\theta$$
$$p_y = \tau_{xy} \cos\theta + \sigma_y \sin\theta \qquad (1)$$

P_x & P_y are the components of stress resultant acting on AB.

$$\sigma_{x'} = p_x \cos\theta + p_y \sin\theta$$
$$\tau_{x'y'} = p_y \cos\theta - p_x \sin\theta \qquad (2)$$



Substitution of the stress resultants Eq.(1) into Eq.(2)

$$\sigma_{x'} = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta$$
$$\tau_{x'y'} = \tau_{xy}(\cos^2\theta - \sin^2\theta) + (\sigma_y - \sigma_x)\sin\theta\cos\theta \qquad (3)$$

<=>

$$\boxed{\begin{aligned} \therefore \sigma_{x'} &= \frac{P\cos\theta}{A_{x'}} = \sigma_x \cos^2\theta \\ \tau_{x'y'} &= -\frac{P\sin\theta}{A_{x'}} = -\sigma_x \sin\theta\cos\theta \end{aligned}}$$

▼ $\sigma_{y'}$ may be obtained by substituting $\theta + \pi/2$ for θ in the expression for $\sigma_{x'}$

$$\sigma_{y'} = \sigma_x \underbrace{\sin^2\theta}_{\frac{1}{2}(1-\cos 2\theta)} + \sigma_y \overbrace{\cos^2\theta}^{\frac{1}{2}(1+\cos 2\theta)} - 2\tau_{xy} \overbrace{\sin\theta\cos\theta}^{\frac{1}{2}\sin 2\theta}$$

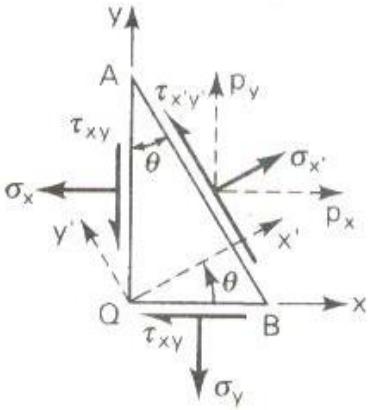
$$\boxed{\sigma_{x'} = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta + 2\tau_{xy} \sin\theta\cos\theta}$$

The transformation equations for stress now become

$$\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau_{xy}\cos 2\theta \qquad (4)$$

$$\sigma_{y'} = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta - \tau_{xy}\sin 2\theta$$



These equations (4) permit the computation of stresses acting on all possible planes AB.

- See Fig.1.11

$$\boxed{\therefore \sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y}$$

2.8 Principal stresses and maximum shear stress in two dimensions

- The orientation of x'y' corresponding to maximum or minimum $\sigma_{x'}$ can be decided using $d\sigma_{x'}/d\theta$.

$$-(\sigma_x - \sigma_y)\sin 2\theta + 2\tau_{xy}\cos 2\theta = 0 \qquad (*)$$

$$\Rightarrow \tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \qquad (**)$$

$$\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

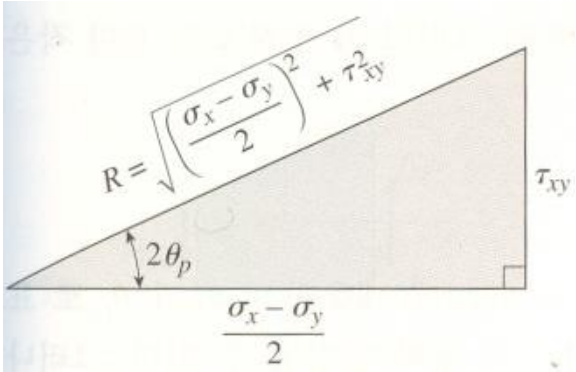
$$\tau_{x'y'} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\sigma_{y'} = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta - \tau_{xy}\sin 2\theta$$

- A principal plane is a x'y' plane of zero shear (from Eqn (*)).

Two directions, $\tan 2\theta = \tan 2(\theta + \pi/2)$, mutually perpendicular, are found to satisfy eqn.(**) (two principal direction)

▼ The principal stresses are determined by substituting Eq(**) into $\sigma_{x'}$, $\sigma_{y'}$ of Eqn.4 in Chap 1.9.



$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \Rightarrow \boxed{\cos 2\theta_p = \frac{\sigma_x - \sigma_y}{2R}} \quad \boxed{\sin 2\theta_p = \frac{\tau_{xy}}{R}}$$

$$\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\Rightarrow \quad \sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 + \sigma_2 = \sigma_x + \sigma_y$$

$$\sigma_2 = \sigma_x + \sigma_y - \sigma_1$$

$$= \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\cos 2(\theta_p + \frac{\pi}{2}) = -\cos 2\theta_p = -\frac{\sigma_x - \sigma_y}{2R}$$

$$\sin 2(\theta_p + \frac{\pi}{2}) = -\sin 2\theta_p = -\frac{\tau_{xy}}{R}$$

=> $$\sigma_{max,min} = \sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2}$$

Two directions, tan2Θ = tan2(Θ+π/2), mutually perpendicular, are found to satisfy eqn.(**) (two principal direction)

- The planes of maximum shearing stress with $d\tau_{x'y'}/d\theta=0$

$$(\sigma_x-\sigma_y)\cos 2\theta+2\tau_{xy}\sin 2\theta=0$$

$$\Rightarrow \tan 2\theta_s=-\frac{\sigma_x-\sigma_y}{2\tau_{xy}} \qquad (***)$$

$$\sigma_{x'}=\frac{1}{2}(\sigma_x+\sigma_y)+\frac{1}{2}(\sigma_x-\sigma_y)\cos 2\theta+\tau_{xy}\sin 2\theta$$

$$\tau_{x'y'}=-\frac{1}{2}(\sigma_x-\sigma_y)\sin 2\theta+\tau_{xy}\cos 2\theta$$

$$\sigma_{y'}=\frac{1}{2}(\sigma_x+\sigma_y)-\frac{1}{2}(\sigma_x-\sigma_y)\cos 2\theta-\tau_{xy}\sin 2\theta$$

From Eqn.(**) & (***), we observe that the planes of maximum shearing stress are inclined at 45° from maximum principal stress ($\theta_s=\theta_p+\pi/4$).

Also, two values of θ_s that are 90° apart are observed.
<See Fig1.12>

$$\tan 2\theta_p=\frac{2\tau_{xy}}{\sigma_x-\sigma_y} \qquad (**)$$

- From Eqn.(***) & $\tau_{x'y'}$ of Eqn.4 in chap 1.9.(Two values of θ_s that are 90° apart)

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}} \quad (***)$$

=>

$$\sin 2\theta_s = -\frac{\sigma_x - \sigma_y}{2R}$$

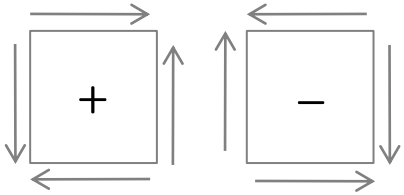
$$\cos 2\theta_s = \frac{\tau_{xy}}{R}$$

$$\sin 2(\theta_s + \frac{\pi}{2}) = \frac{\sigma_x - \sigma_y}{2R}$$

$$\cos 2(\theta_s + \frac{\pi}{2}) = -\frac{\tau_{xy}}{R}$$

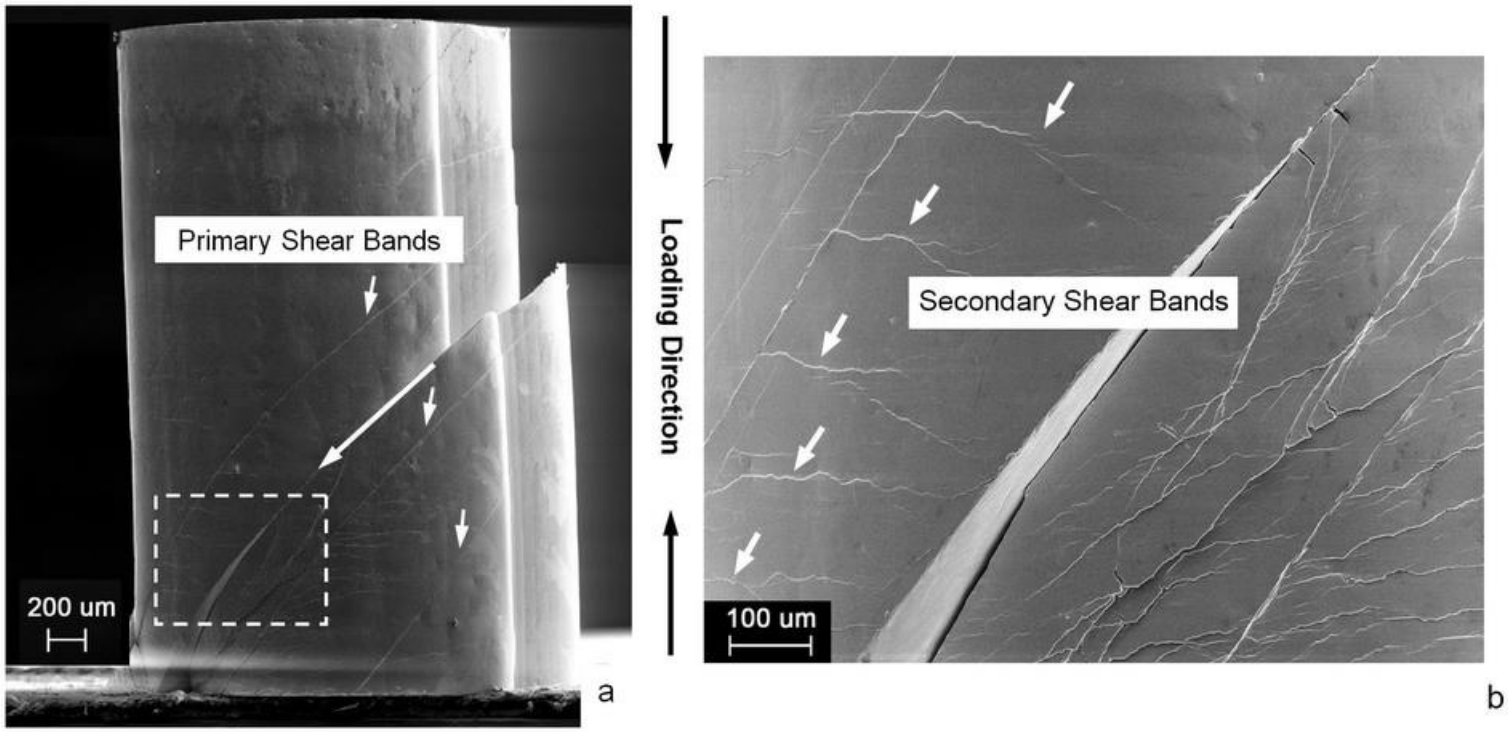
$$\tau_{x'y'} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\Rightarrow \tau_{max} = \pm\sqrt{\left(\frac{\sigma_x - \sigma_y}{2\tau_{xy}}\right)^2 + \tau_{xy}^2} = \pm\frac{1}{2}(\sigma_1 - \sigma_2)$$



- Normal stresses acting on the planes of maximum shearing stresses are obtained using Eqn.(***) & $\sigma_{x'}$ of Eqn.4.

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y)$$



Fracture from shear stress

2.9 Mohr's circle for two-dimensional stress

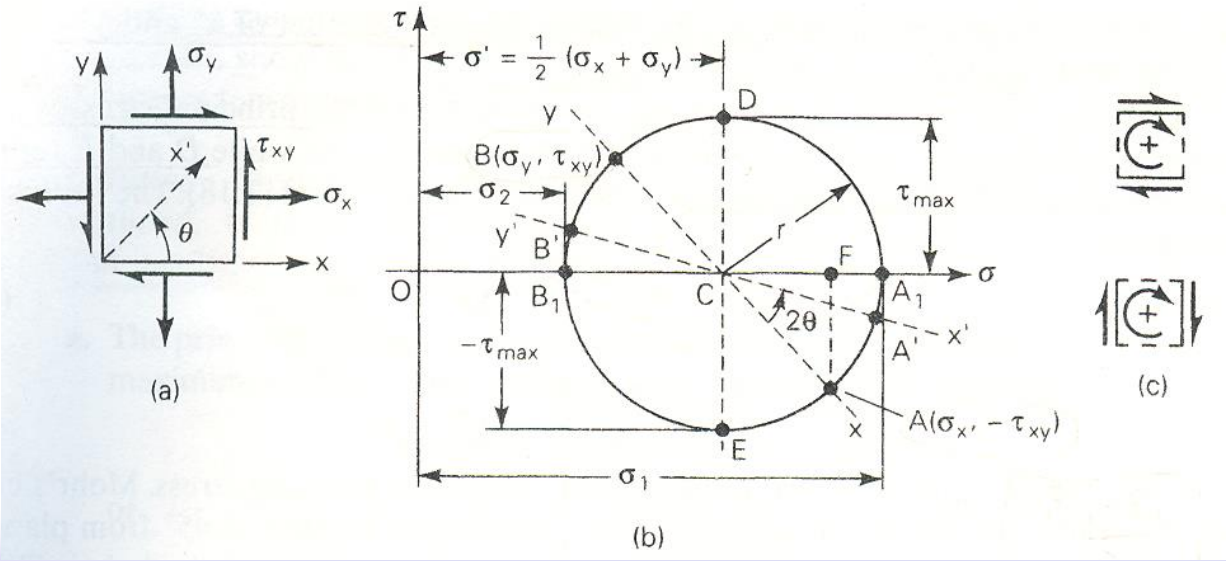
- Mohr's circle permits the rapid transformation of stress from one plane to another and leads also to the determination of the maximum normal & shear stresses.
- Sign convention for shear stress using for Mohr's circle.

Clockwise couple

- The procedure for obtaining Mohr's circle using given σ_x, σ_y & τ_{xy}. (See Fig.1.13)

① Establish a rectangular coordinate system, indicating $+\tau$ & $+\sigma$.

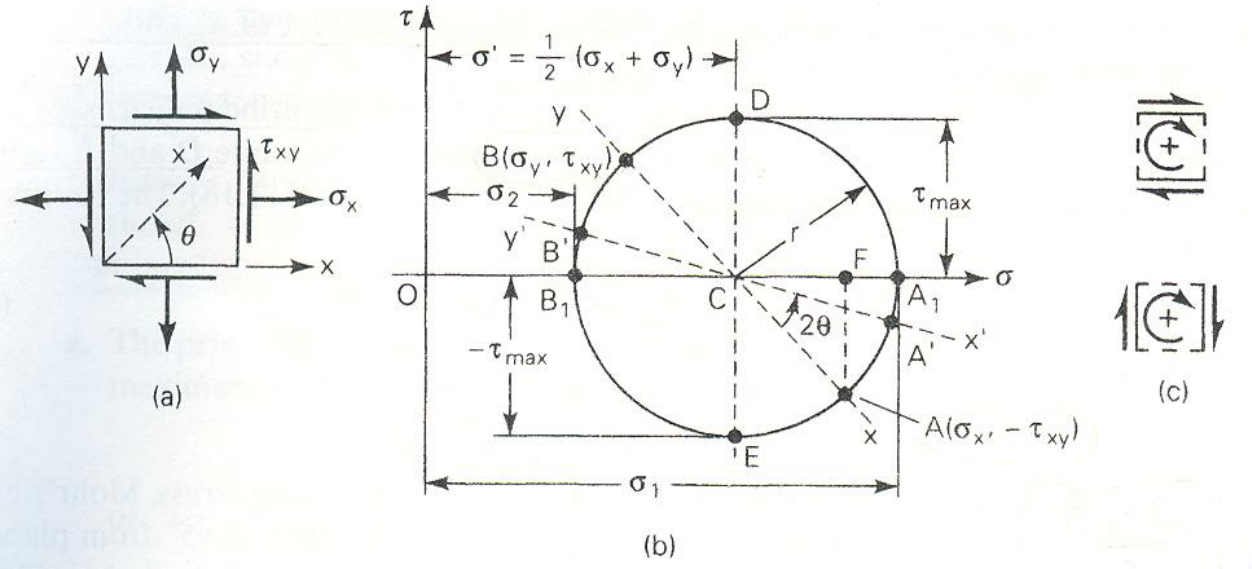
② <u>Locate the center C</u> of the circle on the horizontal axis a distance $(\sigma_x + \sigma_y)/2$ from the origin.



③ Locate point A by coordinates σ_x and $-\tau_{xy}$.
Locate point B by coordinates σ_y and τ_{xy}.

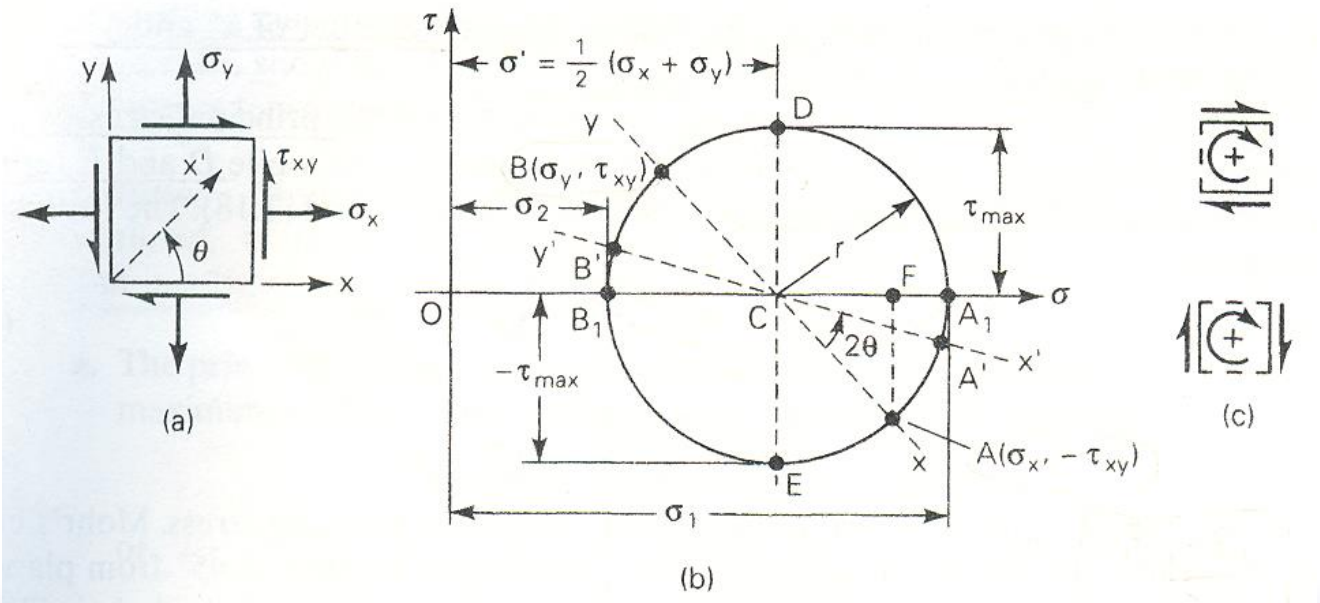
④ Draw a circle with center at C and of radius equal to CA.

⑤ Draw line AB through C.

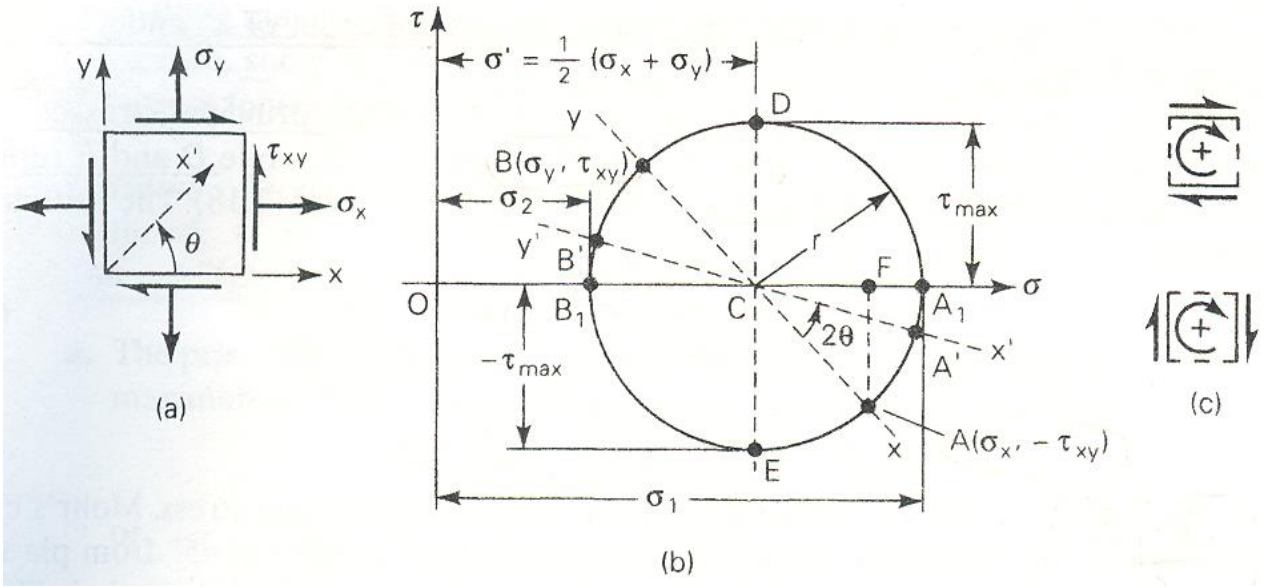


- An angle of 2θ on the circle corresponds to an angle of θ on the element.

- A'B' define states of stress with respect to any other set of x' and y' planes.

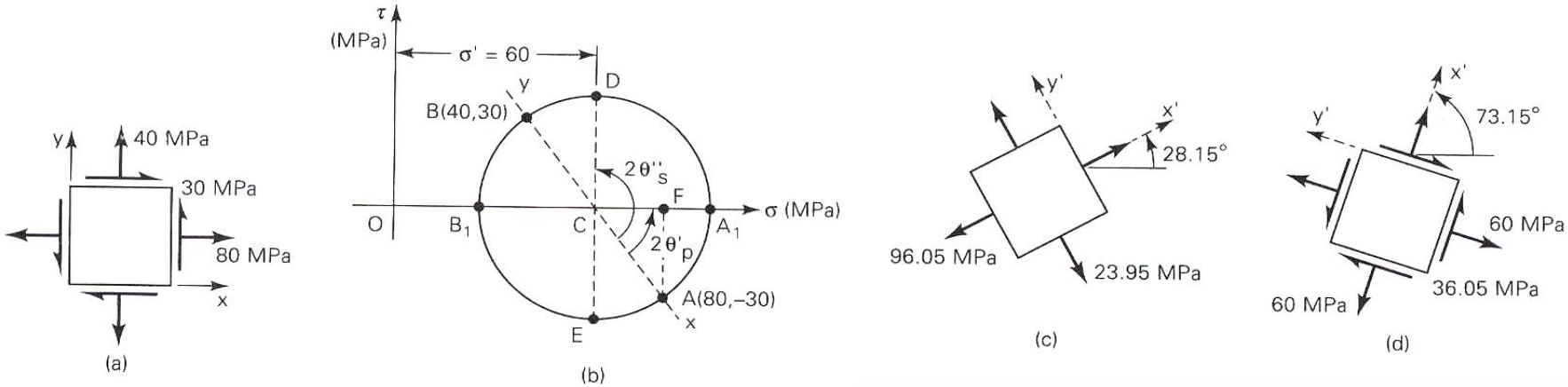


- A_1 and B_1 are the principal stresses.
- The radius of the circle is $CA = \sqrt{CF^2 + AF^2}$ where, $CF=(\sigma_x-\sigma_y)/2$, $AF=\tau_{xy}$



<Solve Example 1.3>

At a point in the structural member, the stresses are represented as in Fig.1.14a.

(a) the magnitude and orientation of the principal stresses.

(b) the magnitude and orientation the maximum shearing stresses and associated normal stresses.